# AI CAREER FOR WOMEN

## Engineering Hub

# Wikipedia Attention Shock Forecaster Using Time-Series Analysis

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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# ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our guide, Mr. Aditya Ardak, for his valuable guidance, support, and encouragement throughout the development of this project.

We also thank AICW Engineering Hub for providing us with this opportunity to work on a real-world problem and enhance our technical skills.

We are grateful to our college and faculty members for their continuous support and motivation. Lastly, we would like to thank our team members for their cooperation and teamwork in successfully completing this project.

# *ABSTRACT*

This project focuses on analyzing Wikipedia pageview data to understand how public attention changes over time. A machine learning–based time-series forecasting model is used to predict expected pageviews for Wikipedia articles. When actual pageviews deviate significantly from predicted values, these are identified as attention shocks, indicating real-world events or trending topics.

The system collects data using the Wikimedia API, preprocesses it, and applies forecasting techniques such as ARIMA or SARIMAX. Residual analysis is used to detect anomalies. The results are visualized using an interactive Streamlit dashboard.

This project helps in identifying patterns in public interest and provides meaningful insights through visual analytics. It can be useful for researchers, analysts, and content creators to understand trends and sudden spikes in attention.

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# CHAPTER 1

# Introduction

**1.1 Problem Statement:**

Wikipedia pageviews are a strong indicator of public interest in real-world topics, events, and personalities. Whenever a major event occurs such as breaking news, viral trends, or important announcements people often visit Wikipedia to learn more, leading to sudden spikes in pageviews. These spikes reflect rapid changes in collective attention, but there is no simple system that can automatically identify and analyze such sudden increases in a structured way.

**1.2 Motivation:**

This project aims to address this problem by developing a system that forecasts expected pageviews using time-series techniques and detects abnormal spikes, known as “attention shocks.” By comparing predicted and actual values, the system highlights unusual patterns and visualizes them through an interactive Streamlit dashboard. This helps researchers, analysts, and content creators understand how public interest changes over time and respond to real-world events more effectively.

**1.3 Objective:**

The main objectives of this project are:

1. To collect Wikipedia pageview data using an API for selected articles and time ranges.
2. To preprocess and organize the data into a proper time-series format by handling missing values and inconsistencies.
3. To develop a forecasting model that can predict expected pageviews based on historical data patterns.
4. To identify and detect abnormal spikes in pageviews, referred to as “attention shocks,” using analytical techniques.
5. To visualize pageview trends and detected shocks through clear and informative graphs.
6. To design and develop an interactive dashboard using Streamlit for easy user interaction and analysis.

**1.4 Scope of the Project:**

1. Focus on Wikipedia data only
2. Uses time-series models
3. Limited to selected articles
4. Works mainly on historical data

# CHAPTER 2

# Literature Survey

## 2.1 Review of Relevant Literature

In recent years, the analysis of online user behavior has become an important research area. Platforms like Wikipedia provide valuable insights into public interest, as users frequently search for information related to current events, trending topics, and popular personalities.

Several studies have explored the use of Wikipedia pageview data to analyze patterns of human attention and information consumption. Researchers have found that sudden increases in pageviews often correspond to real-world events such as breaking news, viral trends, or major announcements.

Time-series analysis has been widely used in such domains to understand temporal patterns in data. Previous work has shown that analyzing historical pageview data can help in identifying trends, seasonality, and anomalies in user behavior. However, many of these studies focus only on data analysis and lack real-time detection and visualization capabilities.

## 2.2 Existing Models, Techniques, and Methodologies

Various models and techniques have been used to analyze and forecast time-series data:

- **Statistical Models:**
  - ARIMA (AutoRegressive Integrated Moving Average)
  - SARIMAX (Seasonal ARIMA with exogenous variables)
  - Exponential Smoothing
    These models are effective in capturing trends and seasonal patterns in time-series data.
- **Machine Learning Models:**
  - Random Forest
  - Regression-based models
    These models help in capturing non-linear relationships in the data.
- **Feature Engineering Techniques:**
  - Lag features (previous day/week values)
  - Rolling mean and rolling standard deviation
    These help in improving prediction accuracy.
- **Anomaly Detection Methods:**
  - Residual analysis (difference between actual and predicted values)
  - Threshold-based detection
    These methods are used to identify abnormal spikes in data.
- **Visualization Tools:**
  - Matplotlib for plotting graphs
  - Streamlit for building interactive dashboards

These techniques are widely used but are often implemented separately rather than in an integrated system

### 2.3 Gaps in Existing Systems and Proposed Solution

Despite the availability of various models and techniques, existing systems have several limitations:

**Limitations:**

- Lack of **automatic detection of sudden spikes (attention shocks)**
- Limited **integration of forecasting and anomaly detection**
- Absence of **interactive and user-friendly dashboards**
- No **customization options** for users (e.g., sensitivity control)
- Most systems are not designed for **real-time analysis**

The proposed system overcomes these limitations by:

- Combining **time-series forecasting with anomaly detection**
- Using **residual-based shock detection** to identify abnormal spikes
- Providing a **Streamlit-based interactive dashboard**
- Allowing users to:
  - Select articles
  - Choose date ranges
  - Adjust shock sensitivity
- Offering **visual insights** such as trends, forecasts, and detected shocks

This integrated approach makes the system more effective, user-friendly, and suitable for analyzing real-world public attention patterns.

# CHAPTER 3

# Proposed Methodology

## 3.1 System Design

The system is designed as an interactive web-based application that forecasts Wikipedia pageviews and detects attention shocks using time-series analysis.

The overall workflow includes:

- User input through dashboard
- Data collection from Wikimedia API
- Data preprocessing and feature engineering
- Model prediction
- Shock detection
- Visualization of results

### 3.1.1 Registration (User Input Module):

In this project, registration refers to user interaction with the system.

- Users select a **Wikipedia article** (dropdown or custom input)
- Choose **date range** for analysis
- Adjust **shock sensitivity level**

This input is collected through a **Streamlit interface**, which acts as the entry point of the system.

### 3.1.2 Recognition (Prediction & Shock Detection Module):

Recognition in this system refers to identifying patterns and detecting attention shocks.

- The system predicts expected pageviews using a trained model
- It compares actual vs predicted values
- Residual values are calculated
- If residual exceeds threshold → **Attention Shock detected**

This module acts as the core intelligence of the system.

## 3.2 Modules Used

The system is divided into multiple functional modules:

### 3.2.1 Data Collection Module:

- Fetches Wikipedia pageview data
- Uses Wikimedia API
- Retrieves daily pageviews for selected articles

### 3.2.2 Data Preprocessing Module:

- Cleans missing or inconsistent data
- Formats timestamps
- Converts data into time-series format

### 3.2.3 Feature Engineering Module:

Creates features like:

- Lag values (lag_1, lag_7)
- Rolling mean (7-day, 30-day)
- Time-based features (month, day)

### 3.2.4 Forecasting Model Module:

- Loads pre-trained machine learning model
- Predicts expected pageviews
- Helps identify normal behaviour patterns

### 3.2.5 Shock Detection Module:

- Calculates residuals (Actual − Predicted)
- Uses threshold-based detection
- Identifies abnormal spikes (attention shocks)

### 3.2.6 Visualization Module:

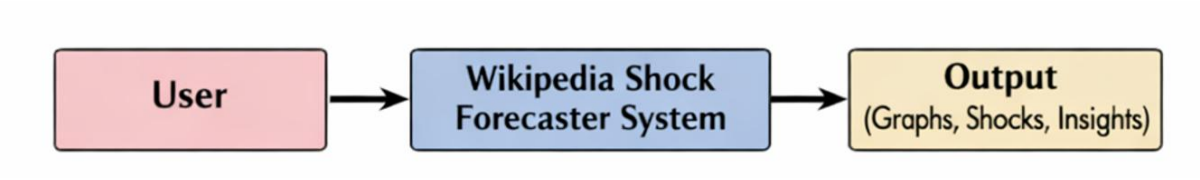
- Displays results using graphs and charts
- Includes:
    - Trend analysis
    - Forecast vs actual comparison
    - Shock detection visualization
    - Insights dashboard

### 3.3 Data Flow Diagram

A Data Flow Diagram (DFD) represents the flow of data within the system and shows how input data is processed to generate output.

It helps in understanding the system structure and interaction between different modules.

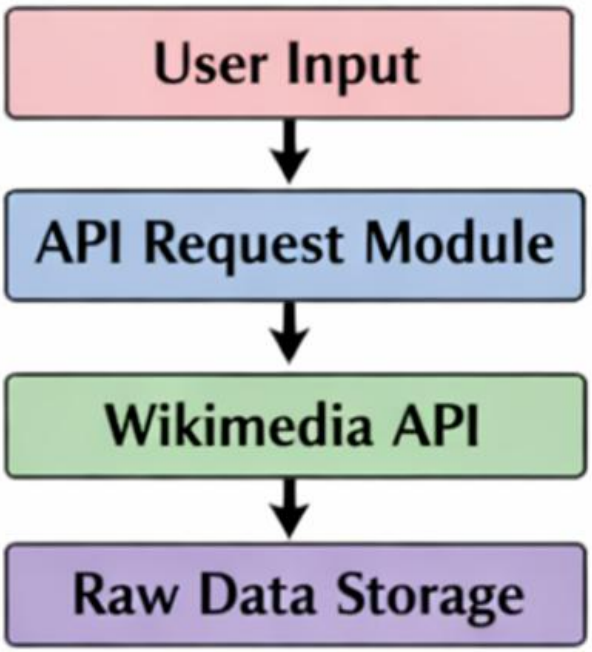
#### 3.3.1. DFD Level 0 (Context Diagram)



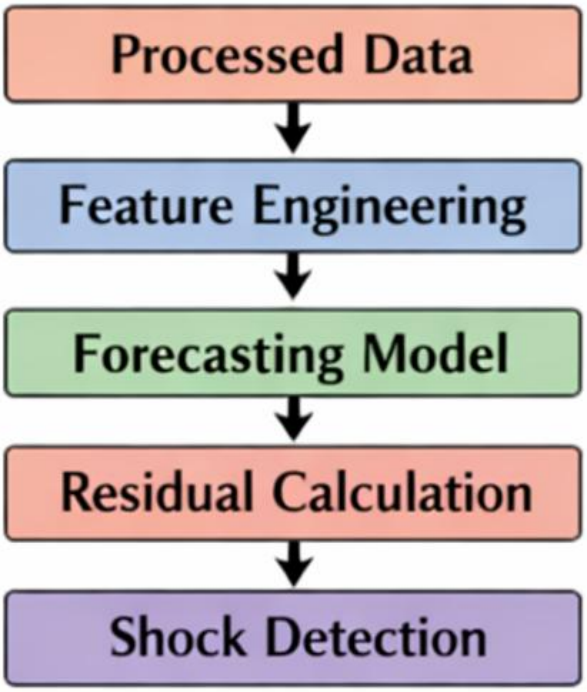

**Description:**

- User provides:
  - Article name
  - Date range
  - Sensitivity
- System processes data and outputs:
  - Trends
  - Predictions
  - Attention shocks

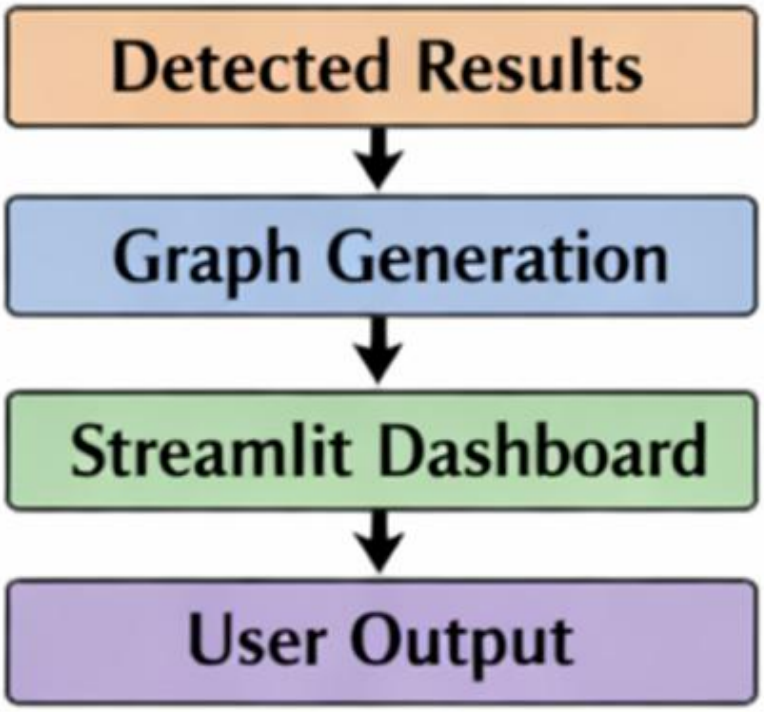
#### 3.3.2. DFD Level 1 - Data Collection Module:



### 3.3.3. DFD Level 1 - Prediction & Shock Detection Module:




### 3.3.4. DFD Level 1 - Visualization Module:



## 3.4 Advantages

- Provides **real-time analysis of public attention**
- Detects **sudden spikes automatically**
- Interactive and **user-friendly dashboard**
- Customizable **shock sensitivity**
- Helps in understanding **real-world events and trends**
- Combines **prediction + anomaly detection + visualization** in one system

## 3.5 Requirement Specification

### 3.5.1. Hardware Requirements:

- Laptop / PC
- Minimum 8 GB RAM
- Processor: Intel i5 or equivalent

### 3.5.2. Software Requirements:

- Operating System: Windows / Linux
- Programming Language: Python
- Development Tools:
  - Jupyter Notebook
  - VS Code
  - Streamlit
- Libraries:
  - Pandas
  - NumPy
  - Statsmodels
  - Matplotlib
  - Plotly

# CHAPTER 4

# Implementation and Result

The system is implemented using **Python and Streamlit** to provide an interactive web-based interface.

The application allows users to:

- Select articles
- Fetch data
- Analyze trends
- Detect attention shocks

## 4.1 System Implementation Overview

The system is implemented using Python and Streamlit to create an interactive web-based dashboard. The application integrates multiple modules including data collection, preprocessing, forecasting, shock detection, and visualization.

The workflow starts with user input (article name, date range, sensitivity), followed by fetching data from the Wikimedia API. The data is then cleaned and transformed into a time-series format. A forecasting model such as SARIMAX is applied to predict expected pageviews, and residual analysis is used to detect attention shocks. Finally, results are displayed through interactive graphs in Streamlit.

## 4.2 Data Collection Implementation

The data is collected using the Wikimedia Pageviews API. The API retrieves daily pageviews for a selected article based on user-defined parameters such as article name, project (e.g., en.wikipedia.org), and date range.

The API response is received in JSON format and converted into a structured dataset for further processing.

## 4.3 Data Preprocessing Implementation

The raw data is preprocessed to ensure consistency and usability. Missing dates are handled by filling gaps in the time-series. Timestamps are converted into proper date formats.

Additional features such as lag values (previous day and previous week), rolling averages (7-day and 30-day), and time-based features (day, month) are created to improve model performance.

### 4.4 Model Implementation (Forecasting)

A time-series forecasting model such as SARIMAX is used to predict expected pageviews based on historical trends and seasonal patterns. The model is trained on past data and generates future predictions.
The predicted values represent the normal expected behavior of pageviews, which is used as a baseline for detecting anomalies.

### 4.5 Shock Detection Implementation

Attention shocks are detected using residual analysis. Residuals are calculated as the difference between actual and predicted pageviews.
A threshold-based method is applied where:

- If residual > threshold → Attention Shock detected
- Otherwise → Normal behavior

The threshold can be adjusted using a sensitivity parameter, allowing users to control how strict the detection should be.

### 4.6 **Streamlit Dashboard Implementation**

The system is deployed using Streamlit, providing an interactive dashboard for users. The interface allows users to:

- Select Wikipedia articles
- Choose date ranges
- Adjust shock sensitivity
- View trends and forecasts

The dashboard displays:

- Time-series graphs
- Forecast vs actual comparison
- Highlighted attention shocks
- Summary insights

### 4.7 **Results and Output**

The system successfully generates meaningful results:

- Accurate prediction of pageview trends
- Detection of sudden spikes (attention shocks)
- Clear visualization of trends and anomalies

The results are displayed in graphical form, where spikes are highlighted clearly. The system is able to identify real-world events reflected as sudden increases in pageviews.

### 4.8 Sample Output Explanation

In the output graph:

- The blue line represents actual pageviews
- The orange line represents predicted values
- Red markers indicate detected attention shocks

These shocks correspond to sudden increases in public interest, often caused by trending events, news, or viral topics.

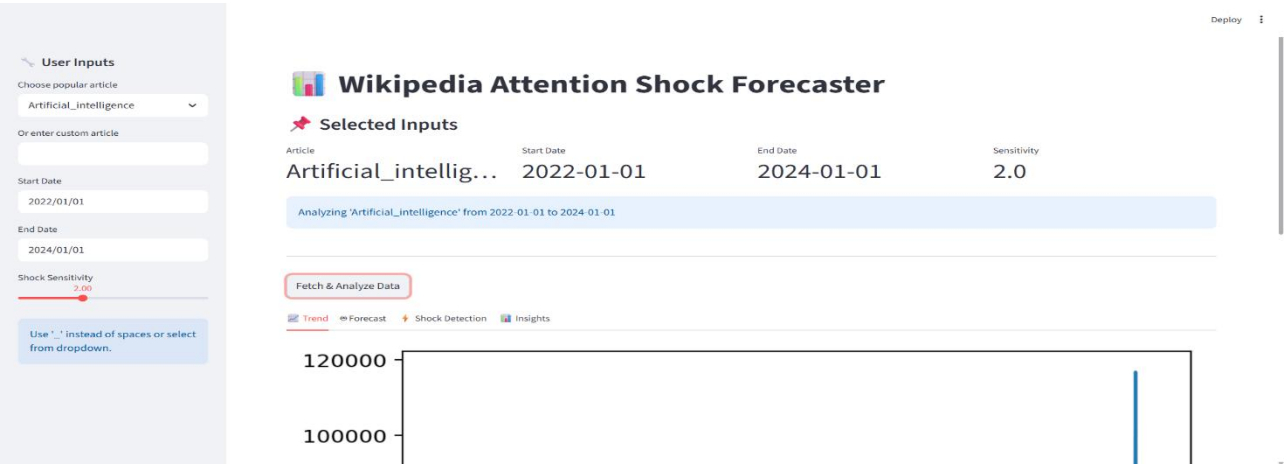



Figure 4.1: Streamlit Dashboard – User Input Interface

- This above figure shows the main dashboard where the user selects the article, date range, and shock sensitivity before starting analysis.
- This graph displays the daily Wikipedia pageviews trend for the selected article over the chosen time period, highlighting variations in public attention.

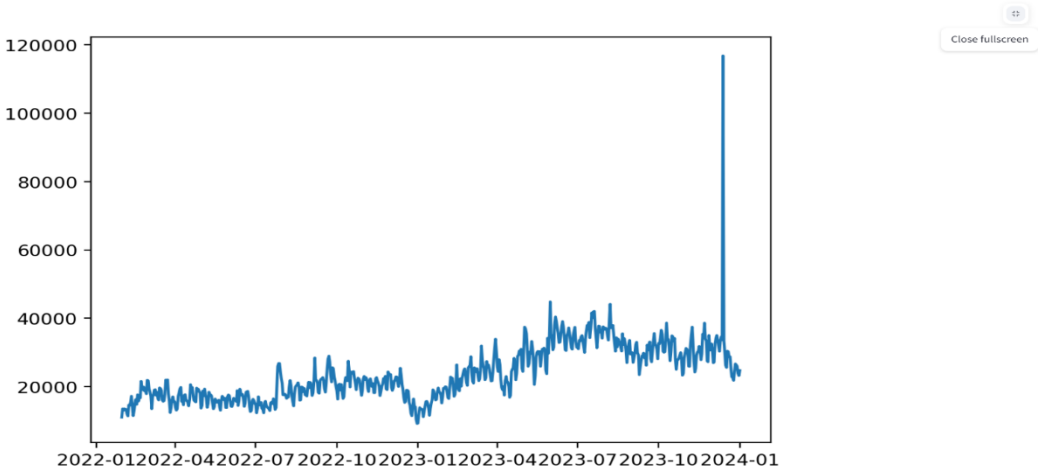



Figure 4.2: Time-Series Trend Visualization

- This graph compares actual pageviews with predicted values generated by the forecasting model, showing how closely the model follows real trends.

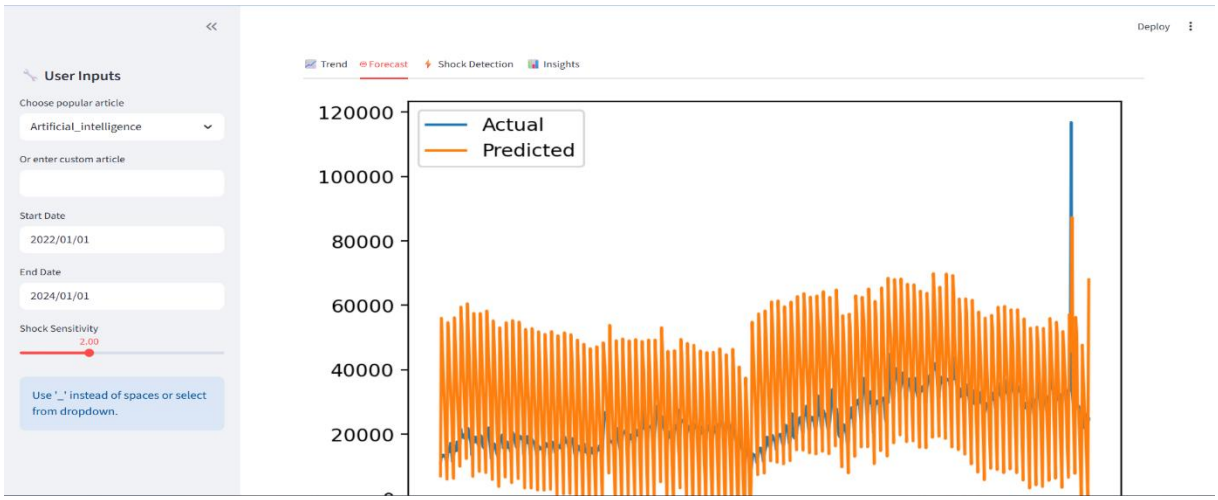



Figure 4.3: Forecast vs Actual Comparison

Figure 4.4: Attention Shock Detection Output

- This above figure shows detected attention shocks along with total shock count and attention status based on residual analysis.
- This visualization highlights a major spike in pageviews, representing a sudden increase in public attention due to real-world events.

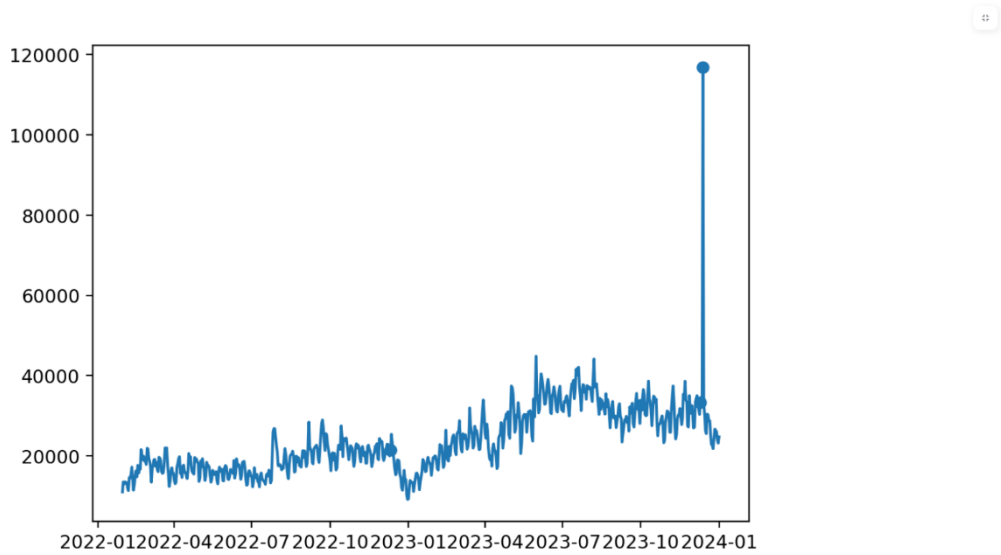



Figure 4.5: Spike Highlight in Time-Series Graph

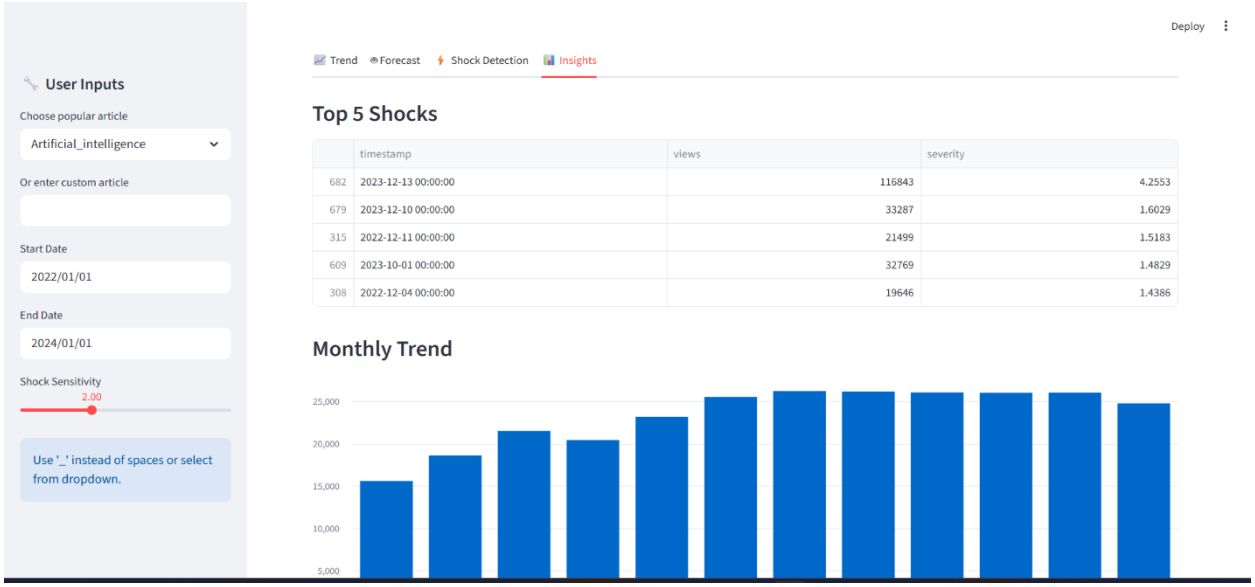



Figure 4.6: Top Detected Shocks Table

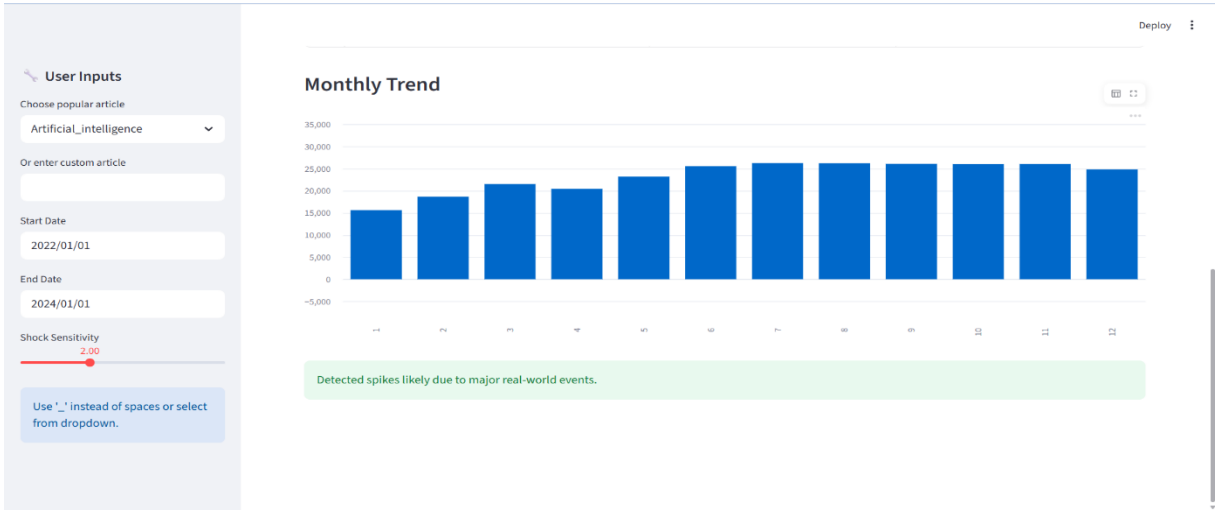



Figure 4.7: Monthly Trend Analysis

# CHAPTER 5

# Discussion and Conclusion

## 5.1 Key Findings:

- The system successfully forecasts Wikipedia pageviews
- Attention shocks are accurately detected
- Interactive dashboard improves usability
- Sensitivity control enhances flexibility

**5.2 Git Hub Link of the Project:** Share the GitHub link

**5.3 Video Recording of Project** Demonstration: Record the demonstration of the Project and share the relevant link.

## 5.4 Limitations:

- Depends on historical data quality
- May miss very small spikes
- Model accuracy varies for unpredictable events

## 5.5 Future Work:

- Real-time streaming data integration
- Use deep learning models (LSTM, GRU)
- Integration with news APIs
- Automated alert system

## 5.6 Conclusion:

The project successfully developed a **Wikipedia Attention Shock Forecaster** using time-series analysis and machine learning techniques.

It effectively predicts expected pageviews and identifies abnormal spikes, providing meaningful insights into public attention trends. The integration of forecasting, anomaly detection, and interactive visualization makes this system a powerful tool for analyzing real-world events and user behavior.

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